

Gurjeet Kaur

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LinkedIn | GitHub | Portfolio

Technical Skills

Languages: Python, SQL, C++, JavaScript, MATLAB
Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformers, RNN
Data & Visualization: Pandas, NumPy, SciPy, PySpark, Matplotlib, Plotly, D3.js, Power BI
Backend & Cloud: FastAPI, Flask, AWS (Lambda, S3, DynamoDB, Textract, SQS, SES), Docker, Kubernetes
Databases & Tools: PostgreSQL, MySQL, MongoDB, Redis, Git, GitHub Actions, Pytest

Experience

- Data Science Intern** May 2026 – Aug 2026
Param Infra Private Limited
 - Designed a SQL-based validation layer cross-referencing invoice records across three construction sites to flag duplicate submissions and repeat contractor billing patterns, preventing an estimated \$45,000+ in duplicate payouts
 - Built a Prophet-based time-series forecasting model in Python over pending bill approvals and payment schedules to generate rolling cash-flow projections, giving stakeholders 2–3 weeks of advance warning on projected shortfalls
 - Instrumented a FastAPI platform with real-time user activity telemetry using Prometheus, using adoption data to guide workflow simplifications that cut manual administrative processing time by 40%
- IT Intern** Feb 2025 – Jul 2025
iEminence Inc.
 - Built an interactive financial analytics dashboard using Python, Streamlit, Pandas, and Matplotlib, reducing reporting cycle time by 30%
 - Automated Excel reporting and validation across 5+ multi-sheet financial datasets using NumPy, SciPy, and XlsxWriter, saving approximately 8 hours of manual work per week
- Data Science Intern** Jun 2024 – Jan 2025
Param Infra Private Limited
 - Built a TF-IDF and XGBoost classifier on incident logs to predict site schedule slips two weeks in advance, achieving 88% recall on a held-out test set
 - Modeled material and delivery trends using moving-average forecasting to automate inventory reorder timing, reducing site stockouts by 25%
 - Developed a predictive model for category-level overspend, enabling proactive budget alerts and reducing overall project costs by 8%
- Data Science Intern** Jan 2022 – Jul 2022
Param Infra Private Limited
 - Developed an automated WhatsApp bot for field staff to submit receipt photos using AWS Textract, increasing daily record submission rates by 45%
 - Built a SQL and Z-score anomaly detection pipeline to flag vendor price variances across sites, reducing overbilling errors by 14%
 - Implemented role-based access control (RBAC) for confidential records across finance, operations, and site-management roles

Projects

- Agentic AI Autonomous Portfolio Rebalancing Platform** | *Python, Pandas, CVXPY, PostgreSQL*
 - Engineered an AI-powered portfolio rebalancing platform with drift monitoring, tax-aware optimization, and explainable trade recommendations, backed by 500+ Pytest tests and CI/CD
- Scaling Laws in Symbolic Music Modeling** | *PyTorch, Transformers, LSTM*
 - Trained Transformer and LSTM models on a 143M-token dataset, achieving 2.22 test loss and 9.20 perplexity while identifying power-law scaling trends
- How NYC Feels: Neighborhood Personality Through Urban Data** | *D3.js, NYU Information Visualization*
 - Built an interactive analytics platform featuring six D3.js visualizations and a weighted recommendation engine across 190+ NYC Neighborhood Tabulation Areas

Education

- New York University** Expected May 2027
M.S. in Computer Science GPA: 3.89/4.0
- Thapar Institute of Engineering and Technology** May 2024
M.Sc. in Mathematics and Computing GPA: 3.79/4.0
- Panjab University** May 2022
B.Sc. in Mathematics, Physics, and Chemistry